Gated counter

Input frequency: fi

Number of counts: NC

$N_C = f_i \cdot T_G$ $\quad f_i = \frac{N_C}{T_G}$ $\quad T_G = \frac{N_C}{f_i}$ $\quad f_G = \frac{f_i}{N_C}$

$N_C = f_i \cdot T_G$ $\quad N_C + 1 = (f_i + df_i) \cdot T_G$ $\quad f_i \cdot T_G + 1 = (f_i + df_i) \cdot T_G$

Resolution:

$f_i + \frac{1}{T_G} = f_i + df_i$ $\quad df_i = \frac{1}{T_G}$ Unit: Hz

$df_{rel} = 100\% \cdot \frac{df_i}{f_i}$ $\quad df_{rel} = 100\% \cdot \frac{1}{f_i \cdot T_G}$ Unit: %